# Azure Key Vault – Hands-On Demo Notes

## 1. What We Will Learn

In this demo, we will learn how to:

1. Create an Azure Key Vault.
2. Understand Key Vault access permissions.
3. Assign roles to a user.
4. Add a Secret to Key Vault.
5. Understand the **Key Vault Secrets Officer** role.
6. Understand the **Key Vault Crypto Officer** role.
7. Generate a Key in Key Vault.

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# 2. Create an Azure Key Vault

Go to the Azure Portal and search for:

**Key Vault**

Click:

**Create**

### Step 1: Select Resource Group

Select the required Resource Group.

### Step 2: Give Key Vault a Name

For example:

```
our-first-keyvault
```

The Key Vault name must follow Azure's naming requirements and must be unique where required.

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# 3. Networking Options

During Key Vault creation, under **Networking**, we can control where the Key Vault can be accessed from.

There are two important options:

### Option 1: All Networks

Access to the Key Vault can be allowed from networks outside a restricted virtual-network configuration.

### Option 2: Selected Networks

Access can be restricted to selected networks, such as specific virtual networks.

For example:

```
Selected Networks
       |
       +---- Virtual Network
       |
       +---- Subnet
```

This helps restrict access to the Key Vault.

### Important

For this demo, we select:

**Allow access from all networks**

We can learn about virtual-network restrictions separately.

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# 4. Create the Key Vault

After configuring the required settings:

```
Review + Create
       ↓
Create
```

After deployment is completed:

```
Go to Resource
```

Now our Key Vault has been created.

# 5. Important Concept – Creating a Key Vault Does NOT Automatically Give You Full Access

This is an important point.

Even if **you created the Key Vault**, you may not automatically have permission to perform every operation on:

- Secrets
- Keys
- Certificates

The required Azure RBAC role must be assigned to the appropriate user, group, or application.

For example:

```
User
   |
   | No required role
   ↓
Key Vault
   |
   X
Cannot perform the required operation
```

After assigning the correct role:

```
User
   |
   | Required RBAC Role
   ↓
Key Vault
   |
   ↓
Can perform permitted operation
```

# 6. Add a Secret

Inside the Key Vault:

```
Key Vault
    ↓
Objects
    ↓
Secrets
```

Click:

**Generate/Import**

### Example Secret

Secret Name:

```
SQLDBConnectionString
```

Secret Value:

```
SQL DB connection string
```

Then click:

**Create**

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# 7. Why Can't We Create the Secret?

If the user does not have the required Key Vault data-plane permission, the operation can fail even though the user created the Key Vault.

Therefore, we need to assign the appropriate RBAC role.

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# 8. Assign Key Vault Secrets Officer Role

Go to:

```
Key Vault
    ↓
Access control (IAM)
```

```
    ↓
Add
    ↓
Add role assignment
```

Search for:

```
Key Vault Secrets Officer
```

Select the role.

Then:

```
Next
    ↓
Select Members
    ↓
Select the required user
    ↓
Review + Assign
```

After the role assignment is successful, the user has the permissions associated with that role.

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# 9. Key Vault Secrets Officer

The **Key Vault Secrets Officer** role provides permissions to perform actions on secrets in the Key Vault.

In simple terms:

```
Key Vault Secrets Officer
          |
          ↓
     Work with Secrets
```

For example, depending on the exact role definition and scope, it allows the assigned identity to manage secrets.

### Important

Having the Secrets Officer role does **not** mean the user automatically has permission to manage cryptographic keys.

This is because **Secrets and Keys have different permissions**.

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# 10. Create the Secret After Assigning the Role

Go back to:

```
Key Vault
   ↓
Objects
   ↓
Secrets
```

Click:

**Generate/Import**

Enter:

```
Name:
SQLDBConnectionString

Value:
SQL DB connection string
```

Click:

**Create**

Now the operation should succeed if the role assignment and scope are correct.

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# 11. Secrets Role vs Keys Role

This is an important interview concept.

Suppose we have:

```
Key Vault
   |
   +---- Secrets
```

```
    |
    +---- Keys
```

Giving a user permission to manage **Secrets** does not automatically give that user permission to manage **Keys**.

For example:

```
Key Vault Secrets Officer
          |
          ↓
       Secrets
         ✓

       Keys
          ✗
```

Therefore, a separate role is required for key management.

---

# 12. Try to Generate a Key

Go to:

```
Key Vault
   ↓
Objects
   ↓
Keys
```

Click:

**Generate/Import**

Try to generate a new key.

If the user does not have the required key-management role, the operation will fail.

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# 13. Assign Key Vault Crypto Officer Role

To manage keys, go to:

```
Key Vault
   ↓
Access control (IAM)
   ↓
Add
   ↓
Add role assignment
```

Search for:

```
Key Vault Crypto Officer
```

Select the role.

Then:

```
Next
   ↓
Select Members
   ↓
Select the required user
   ↓
Review + Assign
```

After the role assignment is successful, the user has the permissions associated with the Crypto Officer role.

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# 14. Key Vault Crypto Officer

The **Key Vault Crypto Officer** role is used to perform actions on cryptographic keys according to the permissions defined by the role.

Simple understanding:

```
Key Vault Crypto Officer
          |
          ↓
   Manage/perform actions
       on Keys
```

**Remember**

```
Secrets → Secrets Officer

Keys → Crypto Officer
```

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# 15. Generate a Key

Now go to:

```
Key Vault
    ↓
Objects
    ↓
Keys
```

Click:

**Generate/Import**

Give the key a name.

Example:

```
StorageAccountEncryptionKey
```

Then click:

**Create**

Now the key is created in Azure Key Vault.

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# 16. Complete Flow

The complete process from this demo is:

```
Create Key Vault
        ↓
```

```
Configure Networking
       ↓
Create Key Vault
       ↓
Check Access
       ↓
Assign Secrets Officer Role
       ↓
Create Secret
       ↓
Assign Crypto Officer Role
       ↓
Create Key
```

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# 17. Important Azure RBAC Concept

Azure Key Vault permissions can be controlled using **Azure Role-Based Access Control (RBAC)**.

The basic idea is:

```
Identity
   +
Role
   +
Scope
   ↓
Permissions
```

For example:

```
User
  +
Key Vault Secrets Officer
  +
Key Vault scope
  ↓
Secret-related permissions
```

Another example:

```
User
  +
Key Vault Crypto Officer
  +
Key Vault scope
  ↓
Key-related permissions
```

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# 18. Real-Time Application Example

Suppose we have an ASP.NET Core application.

The application needs:

```
SQL Connection String
```

Instead of storing it directly in:

```
appsettings.json
```

we can store it in:

```
Azure Key Vault
      |
      ↓
SQLDBConnectionString
```

Then we give the application identity the required permission to access the secret.

For example:

```
ASP.NET Core Web App
        |
        | Managed Identity
        ↓
 Azure Key Vault
        |
```

```
        ↓
SQLDBConnectionString
```

This is much safer than hardcoding sensitive information inside the application.

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# 19. Key Points to Remember

### Point 1

Creating a Key Vault does not necessarily mean you have permission to perform every data operation inside it.

### Point 2

Use **Azure RBAC** to control access.

### Point 3

Secrets and Keys have different permissions.

### Point 4

**Key Vault Secrets Officer** → used for managing secrets.

### Point 5

**Key Vault Crypto Officer** → used for cryptographic key operations/management according to its role permissions.

### Point 6

Always follow the **principle of least privilege**.

Give users/applications only the permissions they actually need.

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# 20. Interview Questions

## Q1. I created a Key Vault. Why can't I create a Secret?

Because creating the Key Vault does not necessarily give your identity the required **data-plane permission** to manage secrets.

You need the appropriate Key Vault RBAC role at the required scope.

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## Q2. Which role is used to manage Secrets?

**Key Vault Secrets Officer**

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## Q3. Which role is used for cryptographic Keys?

**Key Vault Crypto Officer** is a key-related role used for cryptographic key operations/management according to its permissions.

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## Q4. If I have Secrets Officer permission, can I automatically create Keys?

**No.**

Secret permissions and key permissions are separate.

```
Secrets Officer → Secrets

Crypto Officer → Keys
```

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## Q5. What is IAM in Azure?

**IAM = Identity and Access Management**

In Azure, **Access control (IAM)** is used to manage who has access to Azure resources and what actions they are allowed to perform.

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# 21. Quick Revision

```
                  Azure Key Vault
                         |
            +-----------+-----------+
            |                       |
         Secrets                  Keys
```

```
       |                      |
Secrets Officer         Crypto Officer
       |                      |
Connection Strings      Cryptographic Keys
Passwords               Encryption
API Secrets             Decryption
```

### One-minute revision

**Create Key Vault → Assign required RBAC role → Create Secret → Assign key-related role → Create Key**

### Most Important Interview Point

> **Azure Key Vault separates permissions for secrets and keys. Having permission to manage secrets does not automatically give permission to manage keys. Use the appropriate Azure RBAC role and follow least-privilege access.**